J3(O): Construction, the Peirce Split, and Verified Frame Relativity via F4

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Abstract. This series has established, in detail, the algebraic structure of its proposed non-representable "wild side" — but its proposed gauge/matter-side anchor, the exceptional Jordan algebra $J_3(\mathbb{O})$, has until now been cited only in conversation, never built or checked. We construct $J_3(\mathbb{O})$ from scratch and verify it is a genuine Jordan algebra. We then test whether it exhibits the same generic, direction-independent uniformity found everywhere on the octonion side of this series, or genuine structural discrimination. The octonion direction within a given off-diagonal slot does not matter, nor does which of the three off-diagonal slots is chosen — both wash out under real symmetries of the algebra. But the split between diagonal and off-diagonal elements is genuine and robust: a real rank difference (17 versus 18), traced exactly to the classical Peirce decomposition. We then construct $\mathfrak{f}_4$, the 52-dimensional Lie algebra of derivations of $J_3(\mathbb{O})$, from scratch, and use it to show this split is not rigid the way the sedenion structure proved to be: a directly-verified derivation moves a purely diagonal element into off-diagonal territory. The stabilizer of a single idempotent is computed to have dimension exactly 36; the stabilizer of the full diagonal frame, dimension exactly 28 — and both are proven isomorphic to $\mathfrak{so}(9)$ and $\mathfrak{so}(8)$ respectively, by explicit restriction maps: each stabilizer, restricted to a canonical invariant subspace, is verified to consist of antisymmetric matrices, injectively and bracket-compatibly, and therefore to span them exactly. The individual $J_3(\mathbb{O})$ facts realized here are classical — the Peirce decomposition, $\mathfrak{f}_4$ as the derivation algebra (Chevalley–Schafer, 1950), the $\mathrm{Spin}(9)$ isotropy of $\mathbb{OP}^2$ (Borel, 1950). What is new is the complete computational realization of all of it from a raw, validated octonion multiplication table, and the structural contrast this makes possible: $J_3(\mathbb{O})$'s diagonal/off-diagonal split is basis-dependent, not intrinsic — the opposite of what this series proved for the sedenions.

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1. Introduction

Earlier work in this series builds and verifies, in exhaustive detail, the algebraic structure of $\mathbb{O} \cdot e_8$ — the side shown incapable of supporting direct operators — and $\mathbb{O}$ itself, shown throughout to be structurally uniform: no individual direction, and no pairing of directions, distinguishes itself from any other under G_2's transitivity. The gauge/matter-side anchor this series has referenced in discussion — the 27-dimensional exceptional Jordan algebra $J_3(\mathbb{O})$, connected to F_4, E_6, and E_7 — has not, until now, been built or independently checked. This note does so, and asks the natural question: does $J_3(\mathbb{O})$ exhibit the same generic uniformity found everywhere on the octonion side, or does it show real, structural discrimination — and if so, is that discrimination rigid, or basis-dependent?

One point of positioning, stated upfront. The mathematical facts about $J_3(\mathbb{O})$ realized in this note are classical: the Peirce decomposition goes back to the founding Jordan–von Neumann–Wigner formulation, $\mathfrak{f}_4$ as the derivation algebra to Chevalley–Schafer (1950), the $\mathrm{Spin}(9)$ point-stabilizer on $\mathbb{OP}^2$ to Borel (1950), all surveyed in Baez (2002). The contribution here is not their discovery. It is, first, their complete computational realization from a raw, validated octonion multiplication table — every claim rebuilt and checked rather than cited, including explicit isomorphisms rather than invariant-matching — and second, the structural contrast this makes possible with the sedenion side of this series, where the same investigative battery, applied with the same standards, yielded the opposite result: rigidity where $J_3(\mathbb{O})$ shows fluidity. The individual facts are classical; the contrast is this series' own.

2. Construction and verification

Elements of $J_3(\mathbb{O})$ are 3×3 Hermitian matrices with octonion entries, parametrized as (a, b, c, x, y, z) with $a, b, c \in \mathbb{R}$ and $x, y, z \in \mathbb{O}$:

$$X = \begin{pmatrix} a & z & \bar{y} \\ \bar{z} & b & x \\ y & \bar{x} & c \end{pmatrix}.$$

The Jordan product is $X \circ Y = \frac{1}{2}(XY + YX)$, using ordinary matrix multiplication with entries drawn from the validated octonion table used throughout this series.

Verified directly, in order: the Jordan product of two Hermitian matrices remains Hermitian (not automatic with non-associative entries, and checked rather than assumed); the product is commutative; the defining Jordan identity, $(X \circ Y) \circ (X \circ X) = X \circ (Y \circ (X \circ X))$, holds across ten independent random trials; the dimension is 27 exactly, as parametrized.

3. The Peirce split: genuine discrimination, at last

Every test in the companion papers on this series' octonion side — isolated bimodule action, commutativity, the associator (once correctly attributed) — found pure, generic uniformity. The same tests were repeated within $J_3(\mathbb{O})$'s own structure, as an operator $L_X(Y) = X \circ Y$ on the full 27-dimensional space.

Octonion direction within a fixed off-diagonal slot: no discrimination. All seven imaginary directions, placed in the same slot, give identical rank (18) and identical eigenvalue spectrum.

Slot against slot: no discrimination. The same direction, placed in each of the three off-diagonal slots in turn, gives identical spectra — consistent with the natural S_3 symmetry permuting the three off-diagonal blocks.

Diagonal against off-diagonal: genuine, robust discrimination. Each of the three diagonal unit elements gives rank 17; a unit element placed in each of the three off-diagonal slots in turn gives rank 18 — confirmed across all six, not an artifact of one example.

This is the classical Peirce decomposition, and it was independently verified, not merely cited. For a rank-1 idempotent E, $J_3(\mathbb{O})$ splits into eigenspaces of L_E with eigenvalues $1, \frac{1}{2}, 0$, of dimensions $1, 16, 10$ (Baez, *The Octonions*). L_E's rank is therefore $1 + 16 = 17$. The kernel — the 10-dimensional 0-eigenspace — was computed directly for E_1 and found to coincide exactly with the span of the other two diagonal units and their connecting off-diagonal slot, matching the predicted structure precisely.

4. Constructing $\mathfrak{f}_4$

For Jordan algebras, unlike octonions, the commutative product means $L_a = R_a$ always, and the simple commutator $[L_X, L_Y]$ — no more complex combination needed — gives a genuine derivation. This was verified directly (the Leibniz rule with respect to the Jordan product, across eight independent trials) before being trusted further.

All $\binom{27}{2} = 351$ candidate derivations $[L_X, L_Y]$, built from pairs of basis elements, were computed; a maximal linearly independent subset was extracted via greedy rank-increasing selection, yielding exactly 52 generators — matching $\dim \mathfrak{f}_4 = 52$ precisely, on the first attempt.

5. Frame relativity, directly demonstrated

Before computing stabilizers, the qualitative claim motivating this whole direction was tested directly: does $\mathfrak{f}_4$ actually mix diagonal into off-diagonal, or does the Peirce split persist as a rigid structure the way the sedenion split proved to be?

A single verified derivation $[L_X, L_Y]$, for generic X, Y, was applied to E_1 (purely diagonal). **Result: zero diagonal component, substantial nonzero entries in both remaining off-diagonal slots.** The split is not rigid. This stands in direct, structural contrast to the sedenion result established elsewhere in this series: $G_2 \times S_3$'s diagonal action on $\mathbb{S}$ was proven, exhaustively and from first principles, incapable of relating $\mathbb{O}$ to $\mathbb{O}_1$. F_4's action on $J_3(\mathbb{O})$ is the opposite: fluid, not disconnected.

6. The stabilizer subalgebras

Using the full 52-generator basis, two subalgebras were computed directly as null spaces of the appropriate action.

Stabilizer of a single idempotent E_1 (the subspace of $\mathfrak{f}_4$ annihilating E_1 entirely): dimension 36, matching $\mathfrak{so}(9)$, the known isotropy algebra of F_4 acting on the octonionic projective plane $\mathbb{OP}^2$ (Baez).

Stabilizer of the full diagonal frame (the subspace under which E_1, E_2, E_3 all remain diagonal): dimension 28, matching $\mathfrak{so}(8)$.

Both dimensions matched their predictions exactly, on direct computation — not assumed from the literature, derived independently and compared.

7. Beyond dimension: verifying the identification further

Dimension-matching alone does not establish isomorphism. Three further, independent checks were run on both subalgebras.

Closure. Both stabilizer subalgebras were confirmed closed under the bracket — genuine Lie subalgebras, not merely subspaces satisfying a linear condition.

Compactness. The Killing form was computed directly from each subalgebra's own structure constants (obtained via least-squares fit of $[D_i, D_j]$ against the subalgebra's own basis). Both are negative-definite: eigenvalues ranging from -14 to -3.5 for the E_1-stabilizer, confirming a compact Lie algebra, consistent with $\mathfrak{so}(9)$.

Semisimplicity. Both Killing forms are full-rank (36 and 28 respectively) — non-degenerate, and by Cartan's criterion, semisimple.

A further, striking signature for the frame stabilizer: its Killing form is not merely negative-definite but *exactly* proportional to the identity — all 28 eigenvalues equal to -12.0000. This is a distinctive structural fingerprint, consistent with $\mathfrak{so}(8)$'s unusual degree of internal symmetry (triality), rather than a generic feature any 28-dimensional compact semisimple algebra would share.

8. Closing the identification: explicit isomorphisms

The gap named above — matching invariants versus proven isomorphism — closes by constructing the isomorphisms explicitly, with every step verified computationally.

The frame stabilizer is $\mathfrak{so}(8)$. First, a sharpening predicted by the derivation property and confirmed directly for all 28 basis derivations: each annihilates E_1, E_2, E_3 *outright* — $D(E_i) = 0$, not merely "keeps the diagonal diagonal." (Writing $D(E_i)$ diagonal per the defining condition, the idempotent relation $D(E_i) = 2E_i \circ D(E_i)$ forces every component to vanish.) Each such derivation therefore commutes with every L_{E_i} and preserves each joint Peirce component — in particular each off-diagonal slot separately, confirmed directly: all 28 derivations map each slot into itself with zero leakage. Restricting each derivation to the x-slot yields an 8×8 matrix. Verified: all 28 restrictions are antisymmetric; the 28 are linearly independent (rank exactly 28); and restriction respects the bracket. Since the antisymmetric 8×8 matrices form precisely the 28-dimensional Lie algebra $\mathfrak{so}(8)$ itself, an injective, bracket-compatible map into a space of equal dimension is onto: **the restriction is an explicit isomorphism onto $\mathfrak{so}(8)$.** The same holds, verified identically, for the y- and z-slot restrictions — three parallel realizations of one algebra on three 8-dimensional spaces, the concrete face of $\mathfrak{so}(8)$ triality (Baez, §4.2) and the structural source of the scalar Killing form found in §7.

The single-idempotent stabilizer is $\mathfrak{so}(9)$. Every derivation annihilates the algebra unit (from $D(1 \circ 1) = 2D(1)$), so a derivation with $D(E_1) = 0$ also annihilates $E_2 + E_3$ and preserves the 9-dimensional space spanned by $E_2 - E_3$ and the x-slot — the traceless part of E_1's Peirce zero-eigenspace, in which all nine basis elements share the same norm under the trace form. Confirmed directly for all 36 basis derivations: invariance of this 9-space including the vanishing of every leakage component, antisymmetry of all 36 restricted 9×9 matrices, linear independence (rank exactly 36), and bracket compatibility. The antisymmetric 9×9 matrices form exactly the 36-dimensional $\mathfrak{so}(9)$; by the same dimension argument, **the restriction is an explicit isomorphism onto $\mathfrak{so}(9)$.**

A remark on method: an alternative route to the same identification exhibits each frame-stabilizer element as a triality triple (A, B, C) satisfying $A(xy) = B(x)y + xC(y)$. That route reaches the same place but drags in conjugation and ordering conventions tied to the specific Hermitian parametrization — precisely the kind of convention-sensitivity that produced a failed first bracket in this series' TKK construction. The single-slot restriction proof above avoids those conventions entirely while delivering the full isomorphism.

9. Discussion

This series' sedenion work established that $\mathbb{O}$'s wild-side split is rigid: no automorphism of $\mathbb{S}$, continuous or discrete, connects $\mathbb{O}$ to Gresnigt's $\mathbb{O}_1$. $J_3(\mathbb{O})$'s diagonal/off-diagonal split behaves oppositely — genuine, but not rigid: basis-dependent, actively mixed by F_4 , stabilized only by specific, computed subgroups. The natural physical reading, consistent with the original Jordan–von Neumann–Wigner (1934) formulation of quantum observables as a Jordan algebra: diagonal elements correspond to measurable, mutually exclusive states in a chosen frame; off-diagonal elements, to coherences between them; and which decomposition is "diagonal" depends on the frame chosen, exactly as a choice of measurement basis does in ordinary quantum mechanics. Calling this "gauge symmetry" outright would overreach past what has been shown — a Jordan frame's relativity is a kinematic fact about observables and superpositions, not yet a demonstrated dynamical gauge coupling. What has been shown is narrower and solid: the split is real, and it is fluid, in precise, verified contrast to the sedenion side's proven rigidity.

10. Scope

The $\mathfrak{so}(9)/\mathfrak{so}(8)$ identifications rest on the explicit restriction-map isomorphisms of §8 — dimension, closure, compactness, and semisimplicity alone would have constituted strong evidence but not proof. What remains genuinely outside this note's scope, stated precisely: the classical module decompositions $\mathfrak{f}_4 \cong \mathfrak{so}(9) \oplus \mathbf{16}$ and $\mathfrak{f}_4 \cong \mathfrak{so}(8) \oplus \mathbf{8}_V \oplus \mathbf{8}_L \oplus \mathbf{8}_R$ are confirmed here only at the level of dimension arithmetic ($52 - 36 = 16; 52 - 28 = 8 + 8 + 8$) and slot invariance, not as verified decompositions into the named irreducible representations; and whether the three slot-restriction isomorphisms of §8 are related by outer rather than inner automorphisms of $\mathfrak{so}(8)$ — the sharpest form of the triality statement — was not tested.

References

- Companion papers: *The Operator Mismatch*; *The Sedenion Split $\mathbb{S} = \mathbb{O} \oplus \mathbb{O} \cdot e_8$ *; *O and Gresnigt's Octonion Subalgebras Are Not Related by Any Automorphism of S*.

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